

Leaching of ^{15}N -labeled fertilizer nitrate in undisturbed soil columns after simulated heavy rainfall

Martti Esala & Aku Leppänen

To cite this article: Martti Esala & Aku Leppänen (1998) Leaching of ^{15}N -labeled fertilizer nitrate in undisturbed soil columns after simulated heavy rainfall, Communications in Soil Science and Plant Analysis, 29:9-10, 1221-1238, DOI: [10.1080/00103629809370022](https://doi.org/10.1080/00103629809370022)

To link to this article: <https://doi.org/10.1080/00103629809370022>



Published online: 11 Nov 2008.



Submit your article to this journal [↗](#)



Article views: 144



View related articles [↗](#)



Citing articles: 1 View citing articles [↗](#)

Leaching of ^{15}N -Labeled Fertilizer Nitrate in Undisturbed Soil Columns After Simulated Heavy Rainfall

Martti Esala and Aku Leppänen

Agricultural Research Centre, Institute of Crop and Soil Science, FIN-31600 Jokioinen, Finland

ABSTRACT

A laboratory experiment was conducted at $+5^{\circ}\text{C}$ with undisturbed columns (0.19 m i.d., 0.60 m long) of fine sand, heavy clay, and peat to determine the effect of soil and timing of the rainfall event on the amount of fertilizer nitrogen (N) that might be leached after fertilizer application. The ^{15}N -labeled calcium nitrate $[\text{Ca}(\text{NO}_3)_2]$ equal to 120 kg N ha^{-1} was applied 13 or 1 days before adding a total of 70 mm of water. Drainage water was collected through an inlet attached to the bottom of each column until drainage had ceased nine days later. From the sand columns, 27 and 19% of the fertilizer nitrate-nitrogen ($\text{NO}_3\text{-N}$) applied was leached out when the columns were irrigated 1 and 13 days after fertilizer application, respectively. From the three water-permeable clay columns containing earth worm burrows, 23 and 53% of the fertilizer NO_3 applied was leached out when irrigation took place 1 and 13 days later, respectively; the other five clay columns were impermeable. Less than 0.1% of fertilizer NO_3 was leached out from the peat columns with either incubation treatment. After the experiment, the content of fertilizer N was highest at a depth of 0.12-0.24 m in the peat and clay columns, and at a depth 0.24-0.33 m in the sand columns. In the sand and peat columns, less fertilizer N moved down in the column when soil and fertilizer were incubated before irrigation.

It is concluded that in a rainy and cool spring, substantial amounts of fertilizer N can be leached beyond the reach of plant roots and out of the soil profile. The time passing between fertilizer application and rain may reduce the risk of leaching of applied N at low temperatures by diffusion into smaller soil pores rather than by biological immobilization. With the intensity of irrigation applied here, fertilizer N moved down in the sand and clay columns as by-pass flow and hydrodynamic dispersion, whereas in peat columns it tended to move more as a front.

INTRODUCTION

Leaching of N, mainly NO_3 , from cultivated fields to surface and ground waters causes eutrophication, poses a risk to human and animal health, results in economic loss, and reduced soil fertility. High amounts of NO_3 can be leached if a large amount of water percolates through a soil with high NO_3 content. Nitrate may also be transported as surface runoff if water flows on the surface instead of infiltrating into the soil. During the growing period, the amount of water flowing through a soil is often small because of evapotranspiration. The potential water percolation is at its maximum in seasons when the soil is not covered by plants, that is, in spring before plant emergence and in autumn after ploughing (Power and Broadbent, 1989). Fertilizer application at sowing in spring introduces a high content of NO_3 into the soil, and if precipitation is high, there is a risk of large amounts of NO_3 being leached, especially on light textured soils.

A large amount of NO_3 may leach down in the soil profile if water moves rapidly passing the bulk of the soil as by-pass flow (Dekker and Bouma, 1984; Quisenberry et al., 1994). Microbial immobilization, inorganic adsorption and diffusion of NO_3 into smaller soil pores lessen the risk of NO_3 leaching as by-pass flow (Wild, 1972; Shuford et al., 1977; Barraclough et al., 1983; Addiscott, 1991).

There have been many reports on the leaching of other anions using marker chemicals (Germann et al., 1984; Seyfried and Rao, 1987; Elder and Chieng, 1989; Quisenberry et al., 1994) and of NO_3 (Clothier et al., 1988) as by-pass flow in soil columns in the laboratory. Few studies have, however, been conducted of leaching of fertilizer N in circumstances corresponding to field conditions. Dekker and Bouma (1984) noted that, on average, 30% of fertilizer N had leached from soil cores in a field experiment when the cores were irrigated with 8 to 22 mm of water immediately after fertilizer application. Only 8% of the fertilizer N leached out if it was applied to a moist soil, and the cores were irrigated the day after fertilizer application. In a laboratory experiment with undisturbed soil columns, Vanden Heuvel et al. (1991) observed that during 28 days less than 0.06 and 0.5% of NO_3 leached out of intact soil columns when the soil was irrigated with 76 and 152 mm of water, respectively.

Our study sought to establish the proportion of NO_3 applied as N fertilizer that can be leached after fertilizer has been applied to a crop in a simulated rainy

spring, and the impact that the interval between fertilizer application and rain has on the leaching of NO_3^- . The movement of water and ^{15}N -labeled NO_3^- were monitored in undisturbed soil columns of three soil types.

MATERIALS AND METHODS

Soil Columns

Twenty-four virtually undisturbed soil columns (0.19 m i.d. and 0.60 m long) were taken from three soils, fine sand, heavy clay and peat, in Jokioinen, southwestern Finland ($60^\circ 49'\text{N}$, $23^\circ 30'\text{E}$). To take the columns, a 0.65 m deep PVC pipe provided with a cutting tip was forced with a plastic hammer into the soil, the soil outside the column was dug out and the columns was lifted up. Before digging the soil out and lifting the columns from the soil the surface level inside and outside of the column was measured to control that the soil in the column was not compacted during forcing the column into the soil. The columns were stored at $+3^\circ\text{C}$ for no more than 10 days before starting their pretreatment.

The soils had been ploughed in previous autumn to 0.23 m and the columns were taken after spring harrowing to 0.07-0.08 m depth. Selected characteristics of the soils are presented in Table 1. The soils are tentatively classified as Eutric Fluvisol (fine sand), Eutric Cambisol (heavy clay) and Histosol (peat) according

TABLE 1. The pH, organic C, and particle size distribution of soils.

	Depth (m)		
	0-0.24	0.24-0.42	0.42-0.60
Fine sand			
pH _{water}	6.2	6.1	5.8
Org. C %	1.66	1.51	1.38
Particle size composition, %			
<0.002 mm	18.0	18.6	23.3
0.002-0.02mm	7.8	11.2	12.0
0.02-0.2 mm	37.6	36.8	34.2
0.2-2 mm	36.6	33.4	30.5
Heavy clay			
pH _{water}	6.1	5.8	5.8
Org. C %	4.38	2.34	0.95
Particle size composition, %			
<0.002mm	69.6	81.8	86.9
0.002-0.02mm	13.0	6.5	3.5
0.02-0.2 mm	8.0	7.3	8.1
0.2-2 mm	9.4	4.4	1.5
Peat			
pH _{water}	5.3	4.7	4.8
Org. C %	27.98	35.97	29.56

to the FAO classification. Clay layers were encountered in some of the sand columns at depths of 0.24-0.42 m and 0.42-0.60 m. These layers were, however, well structured with numerous of earthworm burrows, and probably did not restrict water flow in the column.

The lower end of each column was prepared by removing about 1 cm of soil with a knife to obtain a broken surface with open pores. The number of earthworm burrows (cylindrical pores larger than 1 mm) on this surface was counted. There was quite a number of cylindrical pores of this size in the peat soil, but they were root channels rather than earthworm burrows. To prevent plugging of the pores at the bottom of the soil column, a plastic screen was placed on the soil surface, and the empty space was filled with quartz sand with a grain size larger than 2 mm. The empty space of about 1 cm was measured to calculate the exact volume of the soil column. A plastic plate was glued (Köratec HF 300 PVC-hart-Kleber transparent glue) to the lower end of the column, and fitted with a brass inlet and plastic pipe to collect the drainage water (Figure 1). The joint between the bottom plate and the PVC pipe was sealed with silicon (Bostik Silicon 2680). The columns were placed in a room at +5°C for the experiment.

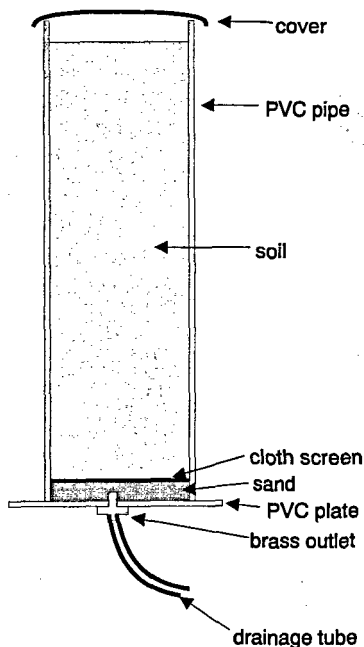


FIGURE 1. The structure of the soil column showing bottom plate with brass outlet to collect drainage water.

To prevent air pockets from forming in the soil, deionized water was applied through the outlet pipe until the water reached the surface. The water was then let out until it had all drained away. This practice was adopted from the method used for measuring water permeability of saturated soils.

Experimental Treatments and Determinations

First, four columns of each soil type were fertilized by injecting ^{15}N -labeled calcium nitrate solution (^{15}N enrichment 9.9486 atom %) into the soil to 5 cm depth. Uniform application was achieved by applying 1 mL of solution with a syringe through each hole of a 19-hole plate. This gave 340.2 mg N per column in 19 mL of water, an amount equivalent to 120 kg N ha⁻¹. Soil samples taken from the same area as the columns were used as control samples for natural abundance of ^{15}N .

Twelve days later, fertilizer was applied to the remaining four columns of each soil in the same way. One day after this application, all the columns were irrigated using an Optifix-dispenser provided with a plastic pipe. Water was added in seven equal doses at one-hour intervals to simulate an average rainfall intensity of 5 mm h⁻¹; and a total of 35 mm. The next day, the columns were irrigated in a similar manner to obtain a total of 70 mm (1984.7 mL) of simulated rainfall 1 and 2 or 13 and 14 days after fertilizer application.

The water draining through the column was collected in plastic bottles. The bottles were changed 1, 2, 3, 6, and 9 days after the start of irrigation. After 9 days, drainage was completed in all the columns and they were transferred to a room at a temperature of +3°C await for sectioning.

After the experiment, the soil columns were sectioned by saw into eight segments: four measuring 0.06 m in the topsoil (0-0.06, 0.06-0.12, 0.12-0.18, and 0.18-0.24 m) and four measuring 0.09 m in the subsoil (0.24-0.33, 0.33-0.42, 0.42-0.51, and 0.51-0.60 m). A 200-cm³ stainless steel cylinder sample was taken from each layer for pore-size distribution determination. The remaining soil in each segment was homogenized. For the determination of total N and ^{15}N -labeled N, a sample was taken, dried at 40°C and finely ground with a Siebtechnik TS 250 disk mill (Siebtechnik GmbH, Germany). For the determination of inorganic N and ^{15}N labeled inorganic N, another sample was taken into a plastic bag and stored frozen at -18°C. The water in the quartz sand at the bottom of the column was collected in plastic bottles for the analysis of inorganic N and ^{15}N labeled inorganic N.

Inorganic forms of N were extracted from the soil with 2M potassium chloride (KCl) (0.100 kg soil, 0.250 L KCl; Keeney and Nelson, 1982; Esala, 1991). The inorganic N [NO_3^- + nitrite (NO_2^-) + ammonium (NH_4^+)] content of the drainage water and soil extracts was determined colorimetrically with a Skalar Autoanalyzer (Henriksen and Selmer-Olsen, 1970; Verdouw et al., 1977; Krom, 1980; Greenberg et al., 1980). The total N and ^{15}N enrichment of the total N in the soil samples

were determined with a VG 622 Micromass mass spectrometer (VG Analytical, UK) linked with a Roboprep-CN-Analyzer (Europa Scientific Ltd, UK; Barrie and Lemley, 1989; Esala, 1991). The inorganic N in both the drainage water and the soil extract was converted to a form suitable for determination of ^{15}N enrichment by applying the diffusion method (Brooks et al., 1989; Esala, 1991).

The pore-size distribution of the soil was determined with the hanging water-column method at potentials -10 kPa (large pores, 30-300 mm) and -1 kPa (macropores, >300 mm) (Jury et al., 1991). The density of the soil solids was measured using a pycnometer (Blake, 1965). The number of earthworm burrows was counted from the lower cut surfaces of the soil of each section.

Statistical Analyses

The significance of differences in the amounts of leached fertilizer N in the drainage water was tested by analysis of variance using the GLM (General Linear Model) procedure of the SAS software (SAS, 1989). The distribution of labeled inorganic N in the soils was examined as an experiment of repeated measurements, in which the measurements were repeated by depths. A linear mixed-effect model was adjusted to the data using the MIXED method of the SAS software. Because some observations were missing, the components of the variances were estimated with the REML method (restricted maximum likelihood; SAS, 1992; Searle et al., 1992). The best covariance structure for the depth factor was estimated with Akaike's information criteria (SAS, 1992).

RESULTS AND DISCUSSION

Inorganic Nitrogen in Drainage Water

Because the columns were brought to field moisture capacity in the pretreatment, practically all the applied water (1984.7 mL) drained through the columns within two days of water addition (Figure 2). Drainage was completed in nine days in all columns. Five of the eight clay columns were essentially impermeable.

Nitrate was the form of N applied and more than 99% of the inorganic N leached was in NO_3 form. Although the ^{15}N enrichment of NO_3 - and NH_4 -N was not determined separately, it can be concluded that NO_3 was the form of labeled N leached. Therefore, the term NO_3 is used later in this paper rather than inorganic N.

The drainage patterns of the sand and peat columns were very similar. In the permeable clay columns drainage was faster than in sand or peat. Leaching of NO_3 was greatest within the first two days after irrigation was started, following the same pattern as drainage. The highest amount of NO_3 , 190 mg corresponding to 67 kg ha^{-1} , leached from the two permeable clay columns irrigated 13 days after fertilizer application (Table 2). No NO_3 was obviously leached from the five impermeable clay columns. Leaching of NO_3 was lowest, corresponding to

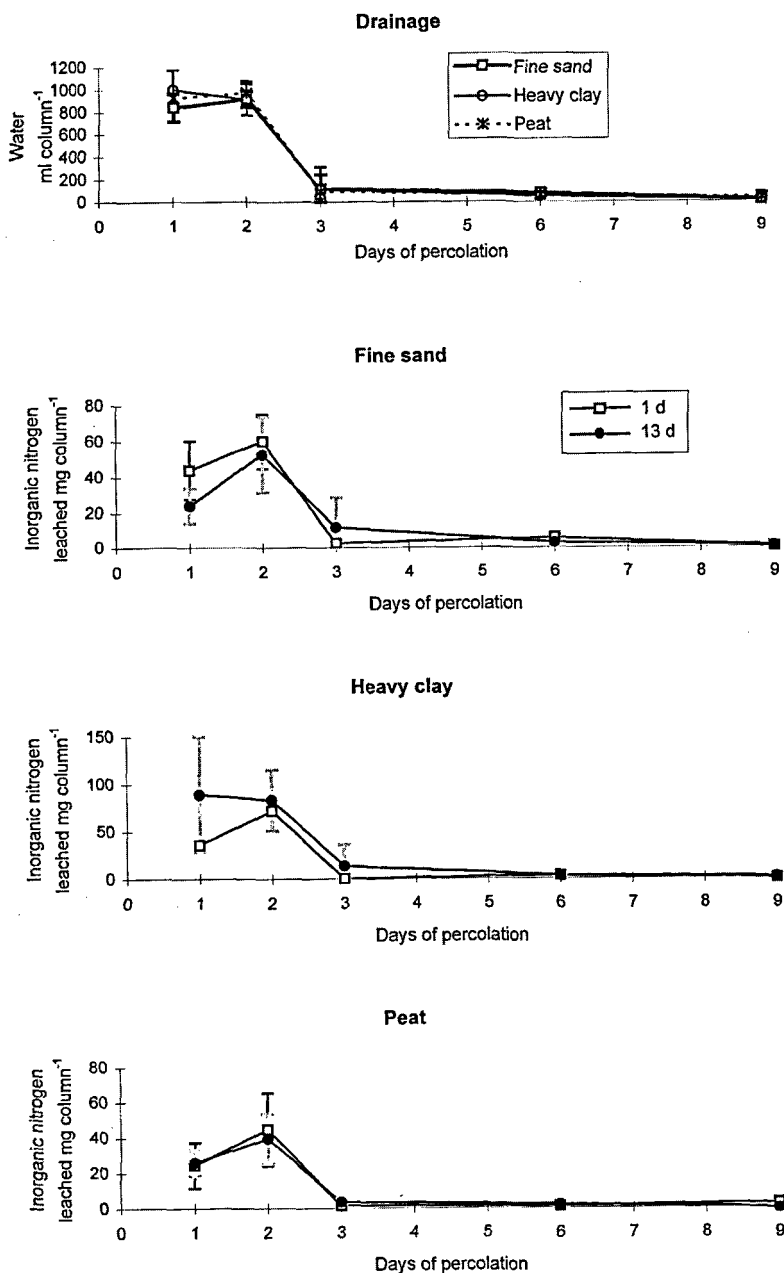


FIGURE 2. Drainage (mL) and average amount (mg) of nitrate leached from columns and the 95% confidence interval.

TABLE 2. Amounts of labeled inorganic nitrogen (kg ha^{-1}) left in columns, and of leached labeled nitrate and the total recovery of labeled nitrogen (%) after irrigating 1 day (1 d) or 13 days (13 d) after fertilizer application.

	Fine sand		Heavy clay		Peat	
	1 d	13 d	1 d	13 d	1 d	13 d
Soil, depth (m)						
0-0.06	2.3	2.4	1.8	1.1	0.8	1.4
0.06-0.12	9.7	9.8	13.9	7.9	16.1	18.3
0.12-0.18	12.2	16.0	19.5	21.9	50.4	39.2
0.18-0.24	11.2	13.8	15.4	12.4	36.2	41.0
0.24-0.33	13.1	16.9	11.6	6.4	12.8	20.4
0.33-0.42	9.5	11.8	7.5	3.2	0.9	1.1
0.42-0.51	5.2	8.8	5.9	0.9	0.1	0.1
0.51-0.60	3.7	4.9	5.0	5.1	0.1	0.1
Sand at bottom	2.0	1.8	1.1	0.8	0.0	0.0
Total in column	68.9	86.2	81.7	59.7	117.4	121.6
Leached ^{15}N labeled nitrate	33.0	22.4	27.1	62.8	0.1	0.0
Soil + leachate	101.9	108.6	108.8	122.5	117.5	121.6
% of labeled nitrate applied (120 kg/ha)	84.9	90.5	90.7	102.1	97.9	101.3

25 kg ha^{-1} , from the peat columns. This finding is in contrast to that of Bergström (1995) who showed that higher amounts of N leach from peat soils than from sand or clay soils. On the other hand, Turtola (1993) and Yläntä et al. (1993) observed that leaching of N in lysimeters was highest from sand and lowest from peat soils.

For the columns irrigated one day following fertilizer application, 27.5, 22.6, and 0.07% of the labeled NO_3 applied leached from sand, clay, and peat columns, respectively (Figure 3). Delaying the water addition for 13 days reduced leaching to 18.7 and 0.02% on sand and peat columns, respectively. More NO_3 (52.6%) was leached from the clay soil incubated for 13 days than from columns incubated for one day. However, there were only three columns that percolated water; two of them were 13 days, and only one 1-day incubation treatments.

Of the total amount of leached NO_3 , 82.3 and 69.5% was derived from the fertilizer for the sand and clay columns, respectively, irrigated one day after fertilizer application. For the columns irrigated 13 days after fertilizer application the figures were 68.5 and 93.7%. The proportion of labeled NO_3 leached was less than 0.01% for the peat soil irrespective of the timing of irrigation relative to fertilizer application. The concentration of labeled NO_3 in the leachate was generally highest two days after the start of irrigation, but it increased slightly in the sand columns (Figure 4).

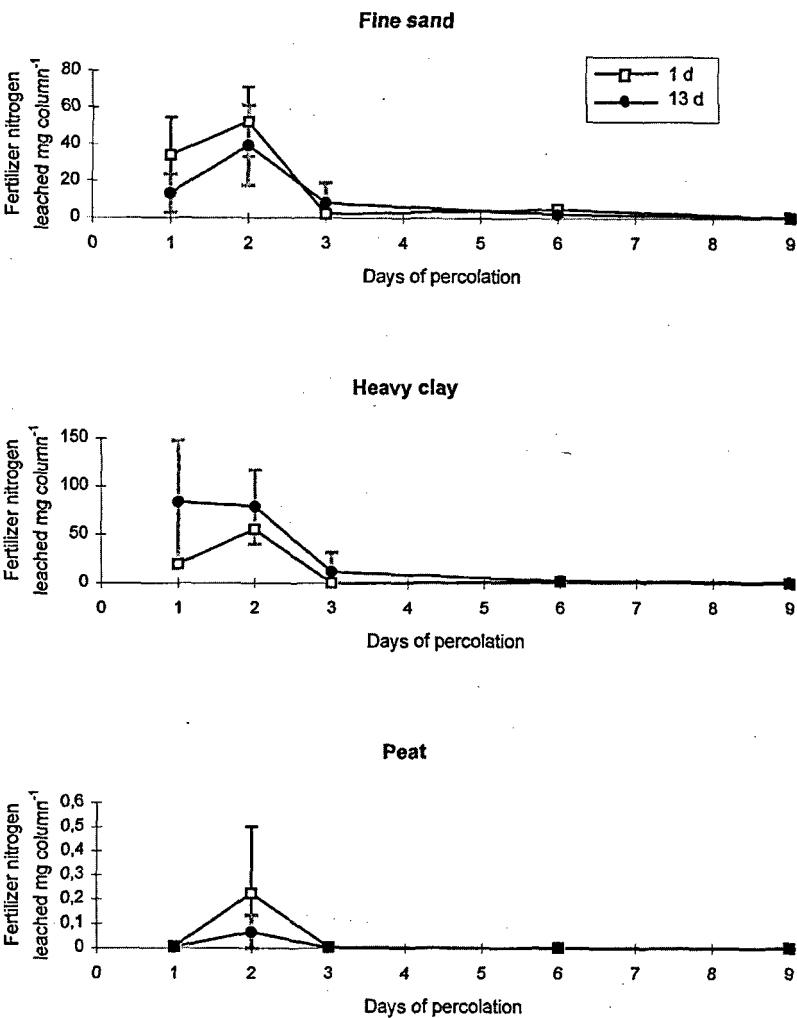


FIGURE 3. Amount of labeled nitrate leached (mg) as average of columns and the 95% confidence interval.

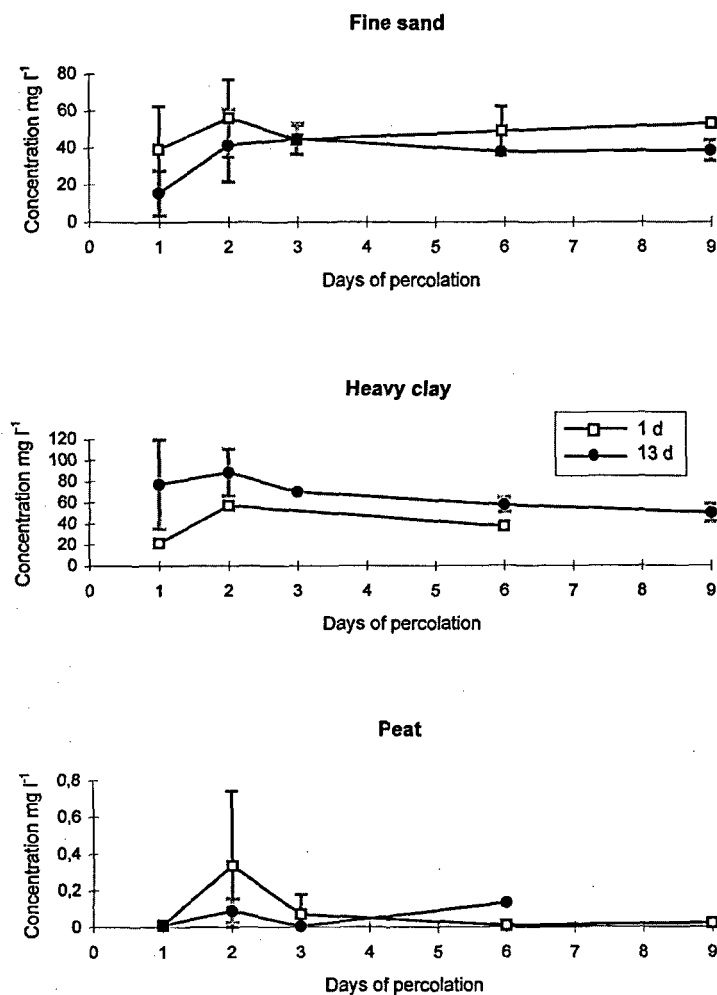


FIGURE 4. Concentration of labeled nitrate in drainage water (mg L⁻¹) and the 95% confidence interval.

Vanden Heuvel et al. (1991) investigated the leaching of NO_3 from 1.10 m deep silty clay lysimeters brought to near-maximum water-holding capacity and fertilized with $168 \text{ kg ha}^{-1} \text{ N}$ as potassium nitrate (K^{15}NO_3). The columns were irrigated three times at one-week intervals at a rate of 0.75 mm h^{-1} . In 28 days, less than 0.06% (0.1 kg ha^{-1}) and 0.5% (0.9 kg ha^{-1}) of the NO_3 added had leached out with irrigations of 76 mm and 152 mm, respectively. Moreover, it took nine days with a higher amount of irrigation for the first fraction of fertilizer to leach from the lysimeters. In contrast, a large amount of fertilizer N was leached out in our experiment and leaching started soon after irrigation. Most of the differences in the results of the two experiments can be explained by the differences in irrigation intensity, temperature ($+5^\circ\text{C}$ versus $+24^\circ\text{C}$), and column depth.

Nitrogen Remaining in Soil

After drainage was completed, the amount of labeled N remaining in the soil was highest for the peat from which practically no labeled NO_3 had been leached (Table 2). The effect of incubation on the labeled N remaining in the clay columns was different from that in the peat or sand columns.

The soil types differed with respect to the depth distribution of labeled N. Less NO_3 was leached down or out from the columns of sandy soil when the water was added 13 days after fertilizer application (Figure 5). During the 13-day period fertilizer NO_3 probably had diffused into smaller pores and so did not move rapidly in the beginning of irrigation with the water flowing in larger pores. Later, as the concentration gradient had changed the fertilizer NO_3 diffused from the smaller pores into water flowing down in the larger pores. This concept is supported by the finding that the concentration of labeled NO_3 in the water during the first days of drainage was lower in the incubated than in the non-incubated columns (Figure 3).

The vertical distribution of labeled total N in the columns was very similar to that of labeled inorganic N (Figure 6). In contrast to the sandy soil in the three out of the eight columns of clay soil, the leaching pattern of NO_3 down in the column was explained by pore-size distribution through the whole column rather than by diffusion of fertilizer NO_3 during incubation. In the impermeable clay columns, there was a compacted layer at 0.33-0.42 m depth that prevented water flowing through the columns.

In the clay and peat soils, the content of labeled N was highest at the 0.12-0.18 m depth, except for the peat soil incubated for 13 days, where the highest content of labeled N was found at 0.18-0.24 m. In the peat columns, labeled N was not found deeper than 0.33 m, and in the sand columns, the content of fertilizer N was highest at 0.24-0.33 m depths.

The residual fertilizer N was distributed more evenly through the sand and clay columns than the peat. This was most likely owing to the greater by-pass flow and hydrodynamic dispersion in the sand and clay than in the peat columns

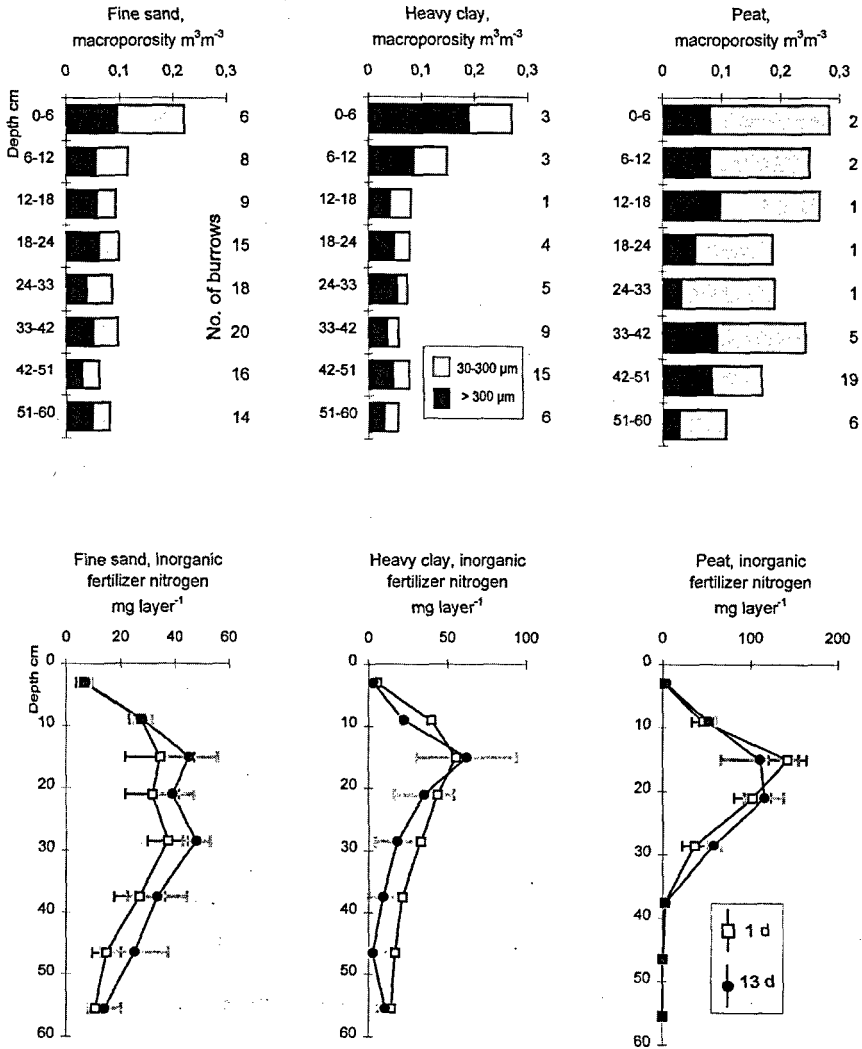


FIGURE 5. Macroporosity (m^3m^{-3} , large pores 30-300 μm and macropores >300 μm), number of earthworm burrows and amount of labeled inorganic nitrogen (mg layer^{-1}) in the columns after draining and the 95% confidence interval.

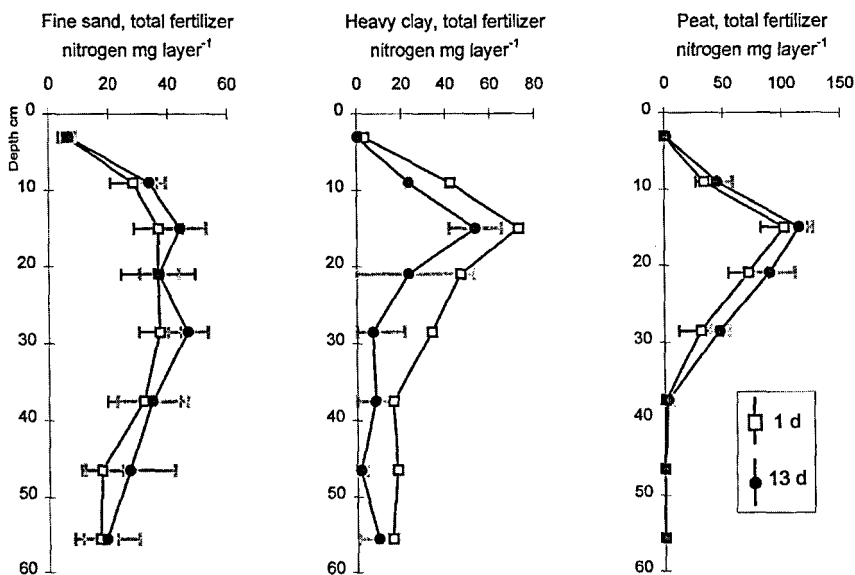


FIGURE 6. Total amount of labeled nitrogen (mg layer^{-1}) in columns after draining and the 95% confidence interval.

where N moved as a more uniform front. The variation in macroporosity between peat columns was large, as their structure was not very uniform. There were no earthworm burrows or other continuous large pores through which by-pass flow could have occurred.

Vanden Heuvel et al. (1991) observed that the largest proportion of N was retained in the upper soil horizons, at 76-mm irrigation at 0-0.35 m, and at 152 mm irrigation at 0-0.65 m. In fact, their results with 76-mm irrigation, which is close to our 70-mm irrigation, are very similar to ours, especially with the peat soil. Only small amounts of N had moved with by-pass flow to lower soil layers.

On the basis of soil moisture, pore volume and bulk density, Vanden Heuvel et al. (1991) calculated that if all the irrigation water had displaced the soil water in one uniform front, the labeled N would have moved to depths of 0.16 m and 0.32 m after 76-mm and 152-mm irrigation, respectively. Calculated on a similar basis, the front of labeled N in our experiment should have moved to depths of 0.25 m, 0.18 m, and 0.15 m in the sand, clay, and peat columns, respectively. In these calculations, it is assumed that irrigation water displaces all the water, even in the smallest pores, which is not realistic. In fact, the residual N content was highest at these calculated depths, except in the peat columns in which the highest contents were found at a 0.18-0.24 m depth.

Other Observations

Because the temperature during incubation was quite low (+5°C), biological immobilization probably did not significantly reduce the leaching of NO₃⁻. At a higher temperature (+24°C), Vanden Heuvel et al. (1991) found that 7-17% of the fertilizer nitrate remaining in the topsoil (0-0.20 m) was immobilized.

The balance of the system shows that 84.9-102.1% of labeled N was accounted for (Table 2). This would suggest some denitrification losses of N especially in the sand columns. However, as the differences between the treatments were not statistically significant and as unaccounted for N is a sum of ten separate determinations, the differences are rather cumulative experimental errors than real losses. Also, denitrification can be assumed to have been minimal at low temperature. In a similar experiment, Vanden Heuvel et al. (1991) measured less than 1 kg ha⁻¹ denitrification, even at higher temperatures.

There was considerable variation between the clay columns in macroporosity and in the number of earthworm burrows (Figure 5). The conductivity of these typical Finnish heavy clay soils decreases abruptly below the top soil, there is a distinct decrease in the macropore volume with depth and the heterogeneity in these soil properties is high (Aura, 1995). To get a more representative sample of this type of clay soils, either the sample diameter or the number of samples should be larger. For sand and peat, the column diameter was sufficient, because the variation, for instance, in water permeability, was rather small. On the other hand, because of the large variation between the clay columns and the existence of impermeable columns, in these soils water and nutrients first move horizontally in the topsoil until they reach a point with permeable subsoil, and then they move to the drainage system or ground water.

The experiment simulates a field situation in which circumstances favour leaching of fertilizer nitrate. However, because there were no significant horizontal movement of water and all the columns could drain freely, the draining of water is not quite comparable to the water flow in a soil. For this reason, both waterflow and NO₃⁻ leaching were likely much faster in the columns than they would have been under field conditions. On the other hand, there are discontinuities of the pores at the lower end of the column at the soil-atmosphere interface where surface tension prevents water from flowing out of the unsaturated soil (Richards et al., 1939; Colman, 1946; Addiscott et al., 1991).

In experiments of this nature, it is possible that some water can flow faster down in the crack between the soil and the column wall. The peat soil in particular is liable to shrink substantially on drying. In our experiment, care was taken to prevent drying of the soils, and they were saturated before the experiment so that they would swell rather than shrink. Contact between the wall and the soil was thus thought to not be an important pathway for water movement. Moreover, the fact that some clay columns were totally impermeable, shows that water did not flow along the pipe wall.

The simulated daily precipitation of 35 mm on two successive days used in our experiment is uncommon in spring in Finland. According to the statistics of 38 weather stations in Finland in 1970-1994, there were only 15 occurrences between May 1 and June 15 when daily precipitation was higher than 35 mm (Anonymous, 1995). However, e.g., in one week, such large precipitation is much more common. Often these occasions raise the question of fertilizer leaching. Also in many other areas such heavy storms are more common.

CONCLUSIONS

Our study shows that considerable amounts of NO_3 can leach to subsoil below the root zone in a cool, rainy spring. There it is not available to plants and NO_3 can more easily be leached further to drainage pipes or ground water with further rain. The high daily precipitations used in our experiment although rare, are not impossible in May or early June which is the time period after sowing and fertilizer applications for most crops in southern Finland. In practice, the amount of fertilizer NO_3 leached would not be expected to be as large as those found in this experiment with the rather shallow soil columns and high precipitation, and in which the soil was water-saturated before N application.

In our experiment, less fertilizer NO_3 was leached from the peat soil than from the sand or clay soils. With the water amounts added, most fertilizer N moved in the sand and clay columns as hydrodynamic dispersion and by-pass flow whereas in the peat soil it tended to move as a front. Examined on a small scale, the movement of water in the large pores determines the movement of NO_3 in sand and clay soils. In the clay soils typical in Finland, the subsoil may be rather impermeable, and water and NO_3 move first as horizontal flow until a crack or a channel in subsoil is reached where the water can penetrate into the lower soil horizons. Had the soil surface been drier, by-pass flow would have been slower until the smaller pores had filled with water. Leaching of NO_3 would then have been slower, and some of the fertilizer NO_3 would have moved into the smaller pores as convection.

The longer the interval between fertilizer application and rain, the smaller is the risk of fertilizer nitrate being leached. In our experiment, the delay reduced the downward movement of fertilizer NO_3 in sand columns, mostly because of NO_3 diffusion into smaller soil pores, and not because of microbial immobilization. Microbial immobilization is more likely to occur at temperatures higher to those in our experiment ($+5^\circ\text{C}$). Diffusion may also be more effective under field conditions, when the soil is usually drier at the time of fertilizer application.

These experiments showed that, in principle, the techniques we applied are suitable for research on the factors and processes affecting NO_3 leaching. However, because of the large variation in the physical properties in clay soils, the diameters of the columns or the number of replicates for the soil type should be larger. Moreover, the irrigation system should be closer to natural precipitation,

and the soil conditions at the beginning of the experiments more like those in the field. Columns deeper than 0.60 m would give results more applicable to field conditions.

ACKNOWLEDGMENTS

We wish to thank J. Ali-Hokka and L. Seppänen for technical assistance. We also thank our colleagues, L. Alakukku, E. Aura, A. Kulmala, J. Pitkänen, and E. Turtola, for their valuable and constructive criticism on the manuscript.

REFERENCES

- Addiscott, T.M., A.P. Whitmore, and D.S. Powlson. 1991. Farming, Fertilizers and the Nitrate Problem. C.A.B. International, Wallingford, United Kingdom.
- Anonymous. 1995. Maatalouden tutkimuskeskuksen säätietojen päivittäisrekisteri. [The Daily Database of Weather Data of the Agricultural Research Centre of Finland].
- Aura, E. 1995. Finite element modeling of subsurface drainage in Finnish heavy clay soils. *Agric. Water Mgt.* 28:35-47.
- Barracclough, D., M.J. Hyden, and G.P. Davies. 1983. Fate of fertilizer nitrogen applied to grassland. I. Field leaching results. *J. Soil Sci.* 34:483-497.
- Barrie, A. and M. Lemley. 1989. Automated $^{15}\text{N}/^{13}\text{C}$ analysis of biological materials. *Am. Lab.* 19:82-91.
- Bergström, L. 1995. Leaching of dichlorprop and nitrate in structured soil. *Environ. Pollution* 87:189-195.
- Blake, G.R. 1965. Particle density. pp. 371-373. In: C.A. Black (ed.), *Methods of Soil Analysis. Part 1. Physical and Mineralogical Properties*. Agronomy 9. American Society of Agronomy, Madison, WI.
- Brooks, P.D., J.M. Stark, B.B. Mcinteer, and T. Preston. 1989. Diffusion method to prepare soil extracts for automated nitrogen-15 analysis. *Soil Sci. Soc. Am. J.* 53:1707-1711.
- Clothier, B.E., T.J. Sauer, and S.R. Green. 1988. The movement of ammonium nitrate into unsaturated soil during unsteady absorption. *Soil Sci. Soc. Am. J.* 52:340-345.
- Colman, E.A. 1946. A laboratory study of lysimeter drainage under controlled soil moisture tension. *Soil Sci.* 62:365-382.
- Dekker, L.W. and J. Bouma. 1984. Nitrogen leaching during sprinkler irrigation of a Dutch clay soil. *Agric. Water Mgt.* 9:37-45.

- Elder, L. and S.T. Chieng. 1989. Effect of water table height and soil physical properties on nutrient leaching. pp. 293-304. In: Toxic Substances in Agricultural Water Supply and Drainage., 2nd Pan-American ICID Regional Conference, 1989, Denver. U.S. Committee on Irrigation and Drainage, Denver, CO.
- Esala, M. 1991. Split application of nitrogen: Effects on the protein in spring wheat and fate of ^{15}N -labeled nitrogen in the soil-plant system. *Annu. Agric. Fenn.* 30:219-309.
- Germann, P., W.M. Edwards, and L.B. Owens. 1984. Profiles of bromide and increased soil moisture after infiltration into soils with macropores. *Soil Sci. Soc. Am. J.* 48:237-244.
- Greenberg, A., J. Connors, and D. Jenkins. 1980. Nitrogen (nitrate). pp. 391-404. In: Standard Methods: For the Examination of Water and Wastewater. 15th ed. American Public Health Association, Washington, DC.
- Henriksen, A. and A.R. Selmer-Olsen. 1970. Automatic methods for determining nitrate and nitrite in water and soil extracts. *Analyst* 95:514-518.
- Jury, W.A., W.R. Gardner, and W.H. Gardner. 1991. Soil Physics. 5th ed. John Wiley & Sons, Inc., New York, NY.
- Keeney, D.R. and D.W. Nelson. 1982. Nitrogen—Inorganic forms. pp. 643-698. In: A.L. Page, R.H. Miller, and D.R. Keeney (eds.), Methods of Soil Analysis. Chemical and Microbiological Properties. Agronomy 9. 2nd ed. American Society of Agronomy, Madison, WI.
- Krom, M.D. 1980. Spectrophotometric determination of ammonia: A study of a modified Berthelot reaction using salicylate and dichloroisocyanurate. *Analyst* 105:305-316.
- Power, J.F. and F.E. Broadbent. 1989. Proper accounting for N in cropping systems. pp. 159-181. In: R.F. Follett (ed.), Nitrogen Management and Ground Water Protection, Developments in Agricultural and Managed-Forest Ecology 21. Elsevier, Amsterdam, The Netherlands.
- Quisenberry, V.L., R.E. Phillips, and J.M. Zeleznik. 1994. Spatial distribution of water and chloride macropore flow in a well-structured soil. *Soil Sci. Soc. Am. J.* 58:1294-1300.
- Richards, L.A., O.R. Neal, and M.B. Russell. 1939. Observations on moisture conditions in lysimeters. II. *Soil Sci. Soc. Am. Proc.* 4:55-59. (Ref. Addiscott et al., 1991).
- SAS. 1989. SAS/STAT User's Guide. Version 6. 4th ed. Volume 2. Statistical Analysis System Institute, Inc., Cary, NC.
- SAS. 1992. SAS Technical Report P-229, SAS/STAT Software: Changes and Enhancements. Release 6.07. Statistical Analysis System Institute, Inc., Cary, NC.

- Searle, S.R., G. Casella, and C.E. McCulloch. 1992. *Variance Components*. John Wiley & Sons, Inc., New York, NY.
- Seyfried, M.S. and P.S.C. Rao. 1987. Solute transport in undisturbed columns of an aggregated tropical soil: Preferential flow effects. *Soil Sci. Soc. Am. J.* 51:1434-1444.
- Shuford, J.W., D.D. Fritton, and D.E. Baker. 1977. Nitrate-nitrogen and chloride movement through undisturbed field soil. *J. Environ. Qual.* 6:255-259.
- Turtola, E. 1993. Phosphorus and nitrogen leaching during set-aside. pp. 207-217. In: *Proceedings of the NJF Seminar No. 228*. Scandinavian Association of Agricultural Scientists, Jokioinen, Finland.
- Vanden Heuvel, R.M., R.G. Hoef, R.L. Mulvaney, and B.R. Montgomery. 1991. Movement of nitrogen-15 labeled nitrate in large undisturbed columns of poorly drained soil. *Commun. Soil Sci. Plant Anal.* 22:809-826.
- Verdouw, H., C.J.A. Vanachteld, and E.M.J. Dekkers. 1977. Ammonia determination based on indophenol formation with sodium salicylate. *Water Res.* 12:399-402.
- Wild, A. 1972. Nitrate leaching under bare fallow at a site in northern Nigeria. *J. Soil Sci.* 23:315-324.
- Ylärinta, T., J. Uusi-Kämpä, and A. Jaakkola. 1993. Leaching of nitrogen in barley, grass ley and fallow lysimeters. *Agric. Sci. Finl.* 2:281-291.